Sourav Danyal

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Summary

Motivated Data Science and AI enthusiast with hands-on experience in building ML models, RAG applications, and API services. Passionate about applying data-driven solutions to real-world problems.

Education

Central University of Himachal Pradesh 2024 – 2026
Master of Computer Applications (MCA)

Government Degree College Shahpur (HPU) 2021 – 2024
B.Sc in Mathematics, Computer Science, Physics

Exams Qualified

- Qualified **UGC NET (Computer Science)**, June 2025 – Eligible for PhD admission.

Projects

- **Multi-Agent AI Research Assistant** Live Demo | GitHub
  Built a full-stack AI research assistant using FastAPI, React, LangGraph, LangChain, Groq LLaMA 3.1, Tavily Search, ChromaDB, and Hugging Face embeddings. Designed a multi-agent workflow with researcher, summarizer, writer, and fact-checker agents to generate sourced, verified research reports. Implemented real-time streaming with Server-Sent Events, document upload support, citation handling, and markdown report generation. Containerized the application with Docker and configured deployment-ready frontend/backend architecture.
- **DocRAG Studio: AI Document Analyzer** Live Demo | GitHub
  Built a production-ready full-stack RAG application using React, FastAPI, LangChain, ChromaDB, and Groq LLaMA 3.1 8B with Hugging Face embeddings for semantic search. Supports context-aware question answering across 5 document formats (PDF, CSV, Excel, TXT, DOCX) with per-file context isolation and source-cited responses. Containerized with Docker and exposed RESTful APIs for scalable document ingestion and querying.
- **Movie Recommendation System** Live Demo | GitHub
  Developed a content-based movie recommendation engine using cosine similarity on the TMDB dataset of 5,000+ films, built with Python, Pandas, and Scikit-learn. Deployed an interactive Streamlit web app with real-time poster and trailer retrieval via TMDB API, cached API calls for performance, and secure environment-based API key management.

Open Source Contributions

- **airblackbox/airblackbox** Merged PR Link
  Added static AI dependency discovery for Python/npm projects with configurable AI-library classification and CycloneDX/SPDX SBOM generation integrated into the `air-blackbox` CLI.
- **langchain-ai/langchain** Merged PR Link
  Fixed `create_agent` model-to-tools routing to prevent `KeyError("model")` when middleware injects synthetic `ToolMessage` objects. Added a regression test covering synthetic tool-message injection through middleware.

Hackathons

- **Meta PyTorch OpenEnv Hackathon** – Finalist. Out of 31,000+ team registrations, our team was among the Top 800 teams advancing to the finale organized at Scaler School of Technology (Bangalore).
- **Tryst (IIT Delhi)** – Secured Second Place in a Mathematical Model competition related to Machine Learning.
- **Chaitanya (HPTU)** – Secured Second Place in a Datathon (Machine Learning).

Technical Skills

- **Programming Languages**: Python, C/C++, Java, SQL
- **Data Science & Generative AI**: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit
- **Frameworks**: TensorFlow, PyTorch, LangChain, FastAPI, Flask
- **Databases**: MySQL, MongoDB, ChromaDB
- **Core Concepts**: OOP, DSA, Operating Systems, DBMS, REST APIs
- **Deployment & Tools**: Docker, Hugging Face Spaces, Git, GitHub, VS Code, Jupyter Notebook, Vercel